

User Interface Design

1

1

Key Definitions

- The user interface defines how the system will interact with external entities

2

2

Key Definitions

- The navigation mechanism provides the way for users to tell the system what to do
- The input mechanism defines the way the system captures information
- The output mechanism defines the way the system provides information to users or other systems

3

3

Interface Design

Typical Design Errors

- lack of consistency
- too much memorization
- no guidance / help
- no context sensitivity
- poor response
- Arcane/unfriendly



4

4

Golden Rules

- Place the user in control
- Reduce the user's memory load
- Make the interface consistent

5

5

Place the User in Control

- Define interaction modes in a way that does not force a user into unnecessary or undesired actions.
- Provide for flexible interaction.
- Allow user interaction to be interruptible and undoable.
- Streamline interaction as skill levels advance and allow the interaction to be customized.
- Hide technical internals from the casual user.
- Design for direct interaction with objects that appear on the screen.

6

6

Reduce the User's Memory Load

- Reduce demand on short-term memory.
- Establish meaningful defaults.
- Define shortcuts that are intuitive.
- The visual layout of the interface should be based on a real world metaphor.
- Disclose information in a progressive fashion.

7

7

Make the Interface Consistent

- Allow the user to put the current task into a meaningful context.
- Maintain consistency across a family of applications.
- If past interactive models have created user expectations, do not make changes unless there is a compelling reason to do so.

8

8

Principles for User Interface Design

- Layout
- Content awareness
- Aesthetics
- User experience
- Consistency
- Minimal user effort

9

9

Layout Concepts

- The screen is often divided into three boxes
 - Navigation area (top)
 - Status area (bottom)
 - Work area (middle)
- Information can be presented in multiple areas
- Like areas should be grouped together

10

10

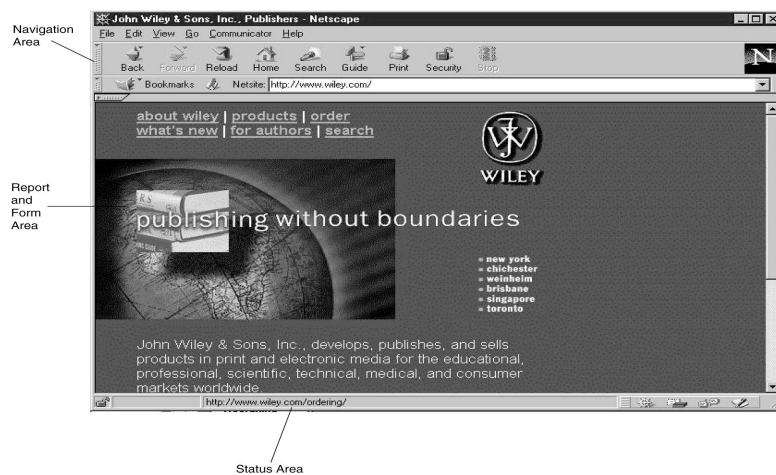
More Layout Concepts

- Areas and information should minimize user movement from one to another
- Ideally, areas will remain consistent in
 - Size
 - Shape
 - Placement for entering data
 - Reports presenting retrieved data

11

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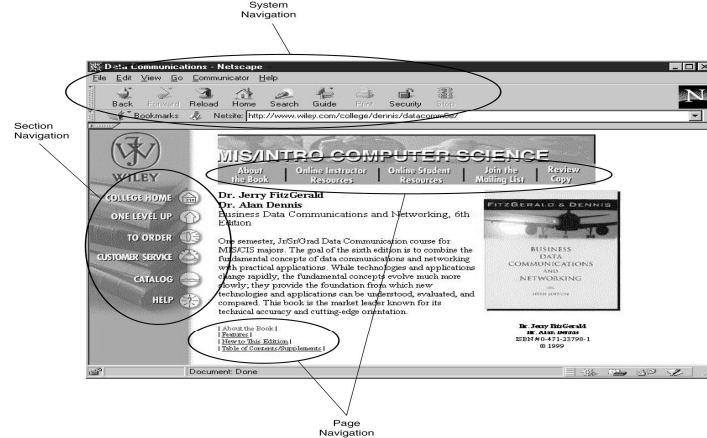
Layout Example 1



12

12

Layout Example 2



13

13

Layout Example 3

Patient Information

Patient Name
 First Name:
 Last Name:

Address:
 Street:
 City:
 State/Province:
 Zip Code/Postal Code:
 Home phone:
 Office Phone:
 Cell Phone:

Referring Doctor:
 First Name:
 Last Name:
 Street:
 City:
 State/Province:
 Zip Code/Postal Code:
 Office Phone:

(a) Vertical flow

Patient Information

Patient Name
 First Name: Last Name:
 Street: City: State/Province: Zip Code/Postal Code:
 Home phone: Office Phone: Cell Phone:

Patient Name
 First Name: Last Name:
 Street: City: State/Province: Zip Code/Postal Code:
 Office Phone:

(b) Horizontal Flow

14

14

Good
flow
in
form

Applicant Information:

Social Security #: Salutation: Current Date:

First Name: Middle Name: Last Name:

Address Line 1: Telephone: Other Information:

Address Line 2:

City: State: Zip Code:

(a) GOOD FLOW

15

15

Bad
flow
in
form

Applicant Information:

Social Security #: Salutation: Current Date: Other Information:

First Name: Last Name: State:

Middle Name: Telephone: Zip Code:

City: Address Line 1:

Address Line 2:

(b) BAD FLOW

16

16

Content Awareness

- All interfaces should have titles
- Menus should show
 - where you are
 - where you came from to get there
- It should be clear what information is within each area
- Fields and field labels should be selected carefully
- Use dates and version numbers to aid system users

17

17

Aesthetics

- Interfaces need to be functional and inviting to use
- Avoid squeezing in too much, particularly for novice users
- Design text carefully
 - Be aware of font and size
 - Avoid using all capital letters

18

18

More Aesthetics

- Colors and patterns should be used carefully
 - Test quality of colors by trying the interface on a black/white monitor
 - Use colors to separate or categorize items

19

19

User Experience

- How easy is the program to learn?
- How easy is the program to use for the expert?
- Consider adding shortcuts for the expert
- Where there is low employee turnover, some training can lessen the impact of less precise interfaces

20

20

Consistency

- Enables users to predict what will happen
- Reduces learning curve
- Considers items within an application and across applications
- Pertains to many different levels
 - Navigation controls
 - Terminology
 - Report and form design

21

21

Minimize Effort

- Three clicks rule
 - Users should be able to go from the start or main menu of a system to the information or action they want in no more than three mouse clicks or three keystrokes

22

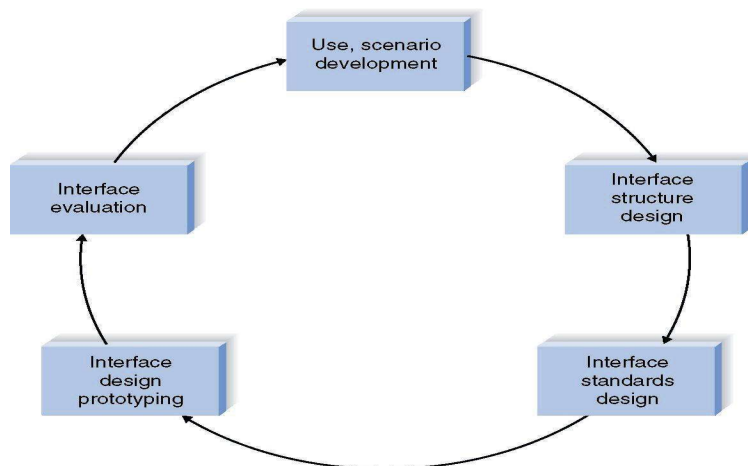
22

USER INTERFACE DESIGN PROCESS

23

23

Five Step Process



24

24

Use Scenario Development

- An outline of steps to perform work
- Presented in a simple narrative tied to the related DFD
- Document the most common paths through the use case so interface designs will be easy to use for those situations

25

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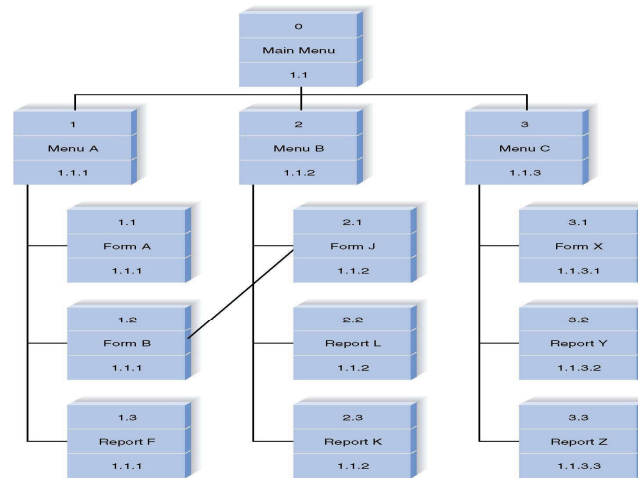
Interface Structure Design

- A diagram showing how all screens, forms, and reports are related
- Shows how user moves from one to another
- Similar to DFD in using boxes and lines
- Boxes denote screens
- Lines show movement from one to another
- Different from DFD in having no standard rules or format

26

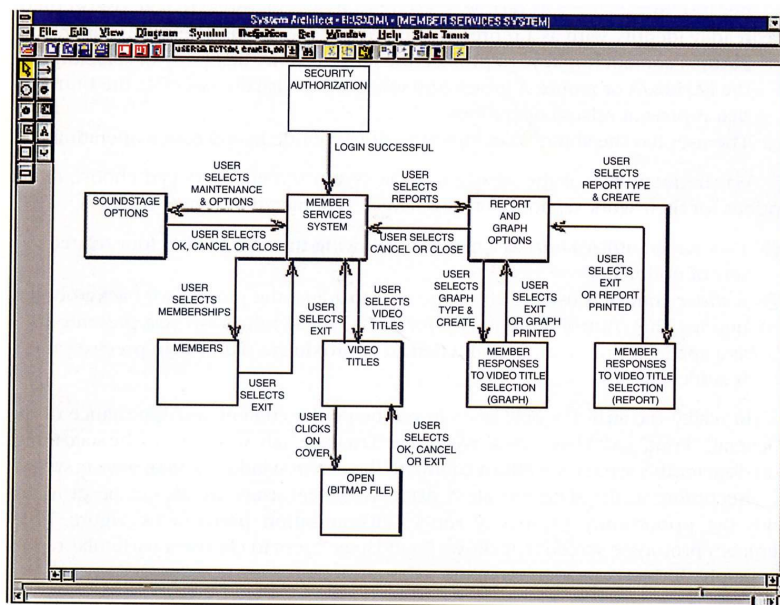
26

Interface Structure Diagram Example



27

27



28

28

Interface Standards Design

- The basic elements that are common across individual screens, forms, and reports within the application
- Interface metaphor
 - Desktop, checkbook, shopping cart
- Interface objects
- Interface actions
- Interface icons
- Interface templates

29

29

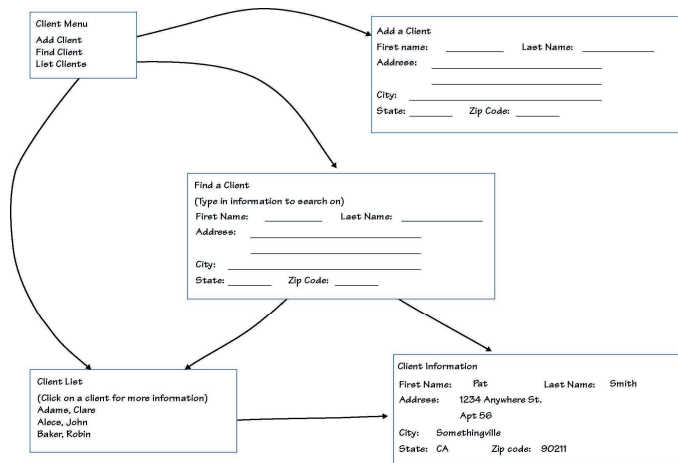
Interface Design Prototyping

- A mock-up or simulation of screen, form, or report
- Common methods include
 - Storyboarding
 - HTML prototype
 - Language prototype

30

30

Storyboard Example



31

31

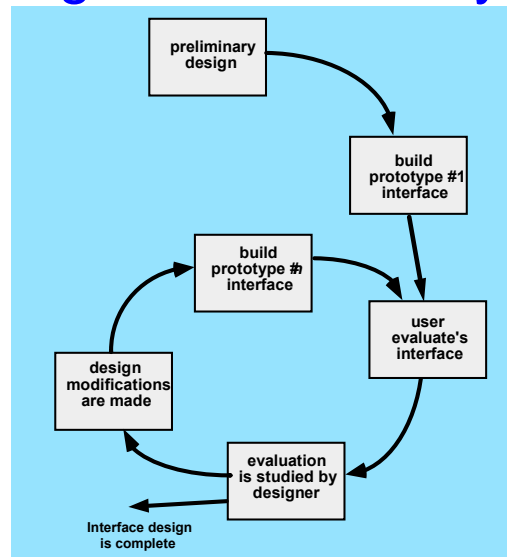
Interface Evaluation Methods

- Heuristic evaluation
 - Compare design to checklist
- Walkthrough evaluation
 - Team simulates movement through components
- Interactive evaluation
 - Users try out the system
- Formal usability testing
 - Expensive
 - Detailed use of special lab testing

32

32

Design Evaluation Cycle



33

33

NAVIGATION DESIGN

34

34

Basic Principles of Navigation Design

- Assume users
 - Have not read the manual
 - Have not attended training
 - Do not have external help readily at hand

35

35

Basic Principles of Navigation Design

- Prevent mistakes
 - Limit choices
 - Never display commands that can't be used (or "gray them out")
 - Confirm actions that are difficult or impossible to undo
- Simplify recovery from mistakes
- Use consistent grammar order

36

36

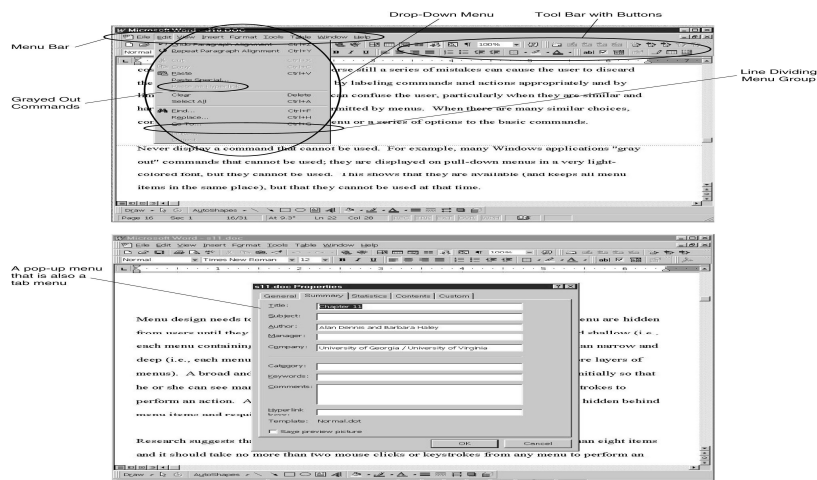
Types of Navigation Control

- Languages
 - Command language
 - Natural language
- Menus
 - Generally aim at broad shallow menu
 - Consider using “hot keys”
- Direct Manipulation
 - Used with icons to start programs
 - Used to shape and size objects

37

37

Common Types of Menus



38

38

Types of Menu

Types of Menu	<i>When Would You Use Each of These Menu Types?</i>
Menu bar Drop-down menu Pop-up menu Tab menu Toolbar Image map	

39

39

Message Tips

- Should be clear, concise, and complete
- Should be grammatically correct and free of jargon and abbreviations (unless they are the users)
- Avoid negatives and humor

40

40

Types of Messages

Types of Messages	<i>When Would You Use Each of These Message Types?</i>
Error message Confirmation message Acknowledgment message Delay message Help message	

41

41

INPUT DESIGN

42

42

Basic Principles of Input Design

- The goal is to simply and easily capture accurate information for the system

43

43

Online versus Batch Processing

- Online processing immediately records the transaction in the appropriate database
- Batch processing collects inputs over time and enters them into the system at one time in a batch

44

44

Capture Data at the Source

- Reduces duplicate work
- Reduces processing time
- Decreases cost
- Decreases probability of error

45

45

Source Data Automation

- Can be obtained by using the following technologies:
 - bar code readers
 - optical character recognition
 - magnetic stripe readers
 - smart cards

46

46

Minimize Keystrokes

- Never ask for information that can be obtained other ways
 - Default values

47

47

Types of Inputs

- Data items linked to fields
- Text
- Numbers
- Selection boxes
 - Check boxes
 - Radio buttons
 - On-screen list boxes
 - Drop-down list boxes
 - Combo boxes
 - Sliders

48

48

Types of Input Forms

The screenshot shows a Netscape browser window displaying a form titled "Sample Input Form". On the left side, labels with arrows point to specific form elements:

- Text Box:** Points to the "Name:" input field.
- Radio Buttons:** Points to the "What is your major:" section, which includes radio buttons for MIS, Accounting, Marketing, Computer Science, and Management.
- Check Boxes:** Points to the "What software do you feel comfortable using:" section, which includes checkboxes for Word, WordPerfect, Excel, Lotus 1-2-3, and Access.
- On-Screen List Box:** Points to the "Select where you were born:" dropdown menu.
- Drop-Down List Box:** Points to the "Hair Color:" dropdown menu.
- Slider:** Points to the "Interest Score:" slider control.

The form also includes a "Document: Done" status bar at the bottom.

49

49

Types of Selection Boxes

Types of Boxes	When Would You Use Each of These Box Types?
- Check box - Radio button - On-screen list box - Drop-down list box - Combo box - Slider	

50

50

Types of Input Validation

Types of Validation	<i>When Would You Use Each of These Validation Methods?</i>
Completeness check Format check Range check Check digit check Consistency check Database checks	

51

51

Input Design Process

1. Identify system inputs and review logical requirements.
 - Determine DFD
 - The data structure for a data flow specifies the attributes or fields to be include in the output.

52

52

Input Design Process (cont'd)

2. Select appropriate GUI controls.
 - Examine the possible values for each attribute to be appear on screen.
3. Design, validate, and test inputs
 - Develop prototype screens for users to review and test.

53

53

Input Design Process

4. If necessary, design the source document
 - If a source document will be used to capture data, it must be designed
 - The source document is for the system user.

54

54

OUTPUT DESIGN

55

55

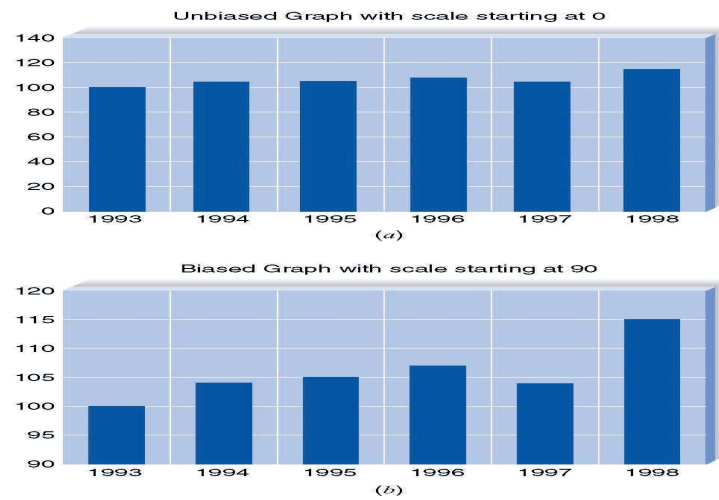
Basic Principles

- Understand report usage
 - Reference or cover-to-cover?
 - Frequency?
 - Real-time or batch reports?
- Manage information load
- Minimize bias

56

56

Bias in Graphs



57

57

Types of Reports

Types of Reports	When Would You Use Each of These Report Types?
Detail reports Summary report Turnaround document Graphs	

58

58

SoundStage Entertainment Club Detailed

PRODUCTS ORDERED ON 1/25/2000

P.O. Number	Product Number	Product Type	Quantity In Stock	Quantity On Order
112312	102774	Merchandise	232	43
	232322	Title	23	43
	232332	Title	2	3
121212	222332	Merchandise	115	132
	546566	Title	667	1
	232554	Title	11,234	343
232323	200992	Title	54,321	1
	1212343	Title	1,324	11
	3434434	Merchandise	6,561	55
	4343434	Merchandise	112	111
	3434344	Title	3	232

Return to Summary Close

Detail report

59

59

SoundStage Entertainment Club Summary

PRODUCT SALES SUMMARY
AS OF 1/25/2000

Product Type	Product Category	Current Month's Unit Sales	Current Year Unit Sales
Merchandise	Clothing	784	4,312
	Media Accessory	541	2,079
	Total:	1,325	6,391
Title	Audio	667	20,439
	Game Title	11,234	12,445
	Video Title	54,321	998,872
	Total:	66,222	1,031,756

View Additional Reports Close

Summary report

60

60

External document



SoundStage Entertainment Club
Fax 317-494-5222

PURCHASE ORDER

The following number must appear on all related correspondence, shipping papers, and invoices:
P.O. NUMBER: 712812

To:
CBS Fox Video Distribution
28253 Rodeo Dr
Hollywood, CA

Ship To:
SoundStage Entertainment Club
Shipping/Receiving Station
Building A
2630 Darwin Drive
Indianapolis, IN 45213

P.O. DATE	REQUISITIONER	SHIP VIA	F.O.B. POINT	TERMS
5-3-06	LDB	UPS		Net 30

QTY	DESCRIPTION	UNIT PRICE	TOTAL
20000	Star Wars: Revenge of the Sith (VHS)	15.99	319,800.00
3000	Star Wars: Revenge of the Sith (DVD Dolby Digital)	19.9	59,970.00
500	Star Wars: Revenge of the Sith (DVD DTS)	24.99	12,495.00
8000	Star Wars: Revenge of the Sith (PlayStation II)	16.99	135,920.00
400	Star Wars: Revenge of the Sith Soundtrack (CD)	16.99	6,796.00
600	Star Wars: Revenge of the Sith Theater Poster	4.99	2,994.00
Subtotal			537,975.00
Tax			37,658.25
Total			575,633.25

1. Please send two copies of your invoice.

2. Enter this order in accordance with the prices, terms, delivery method, and specifications listed above.

3. Please notify us immediately if you are unable to ship as specified.

Madge Worthy
Authorized by

5-4-06
Date

61

61

Tabular report prototype

Microsoft Crystal Reports Professional - [Index Video.rpt]

File Edit View Insert Format Database Report Analyze Window Help

Printed at 2/23/2001
Page Number 1

**Member Response to Video
Title Selection of the Month**

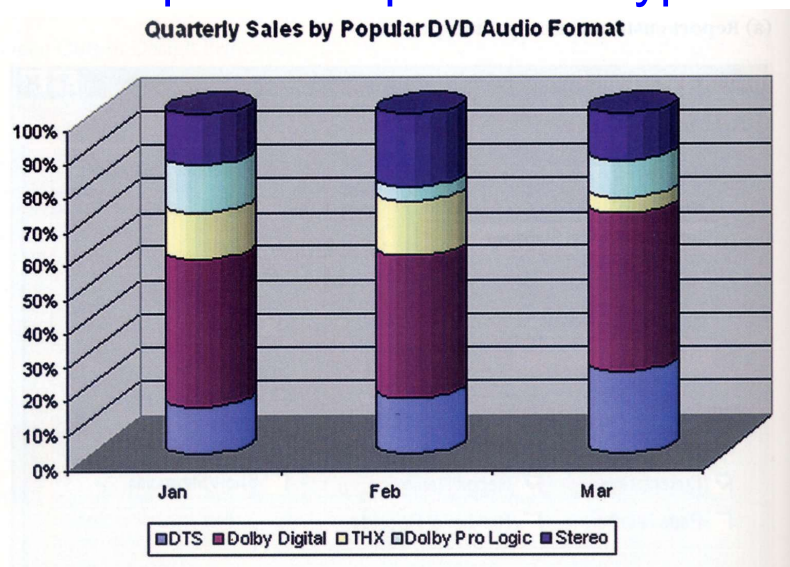


Category	Potential Orders	Selection of the Month	Alternate Selection of the Month + Alternatives	Number of Orders
Action Adventure	6,342	2,410	824	2,867
Animated	3,577	1,536	644	1,241
Comedy	954	181	38	716
Documentary	1,486	877	45	477
Drama	540	389	54	58
Western	104	9	54	40
Horror	920	99	23	2,501
Musical	209	40	78	103
Science Fiction	4,550	2,011	899	5,329
Sports	288	288	277	387
Sum of Potential Orders		19,010		
Sum of Selection of the Month		7,842		
Sum of Alternate Selections		2,916		
Sum of Selection of the Month + Alternatives		3,560		
Total Number of Orders		13,719		

62

62

Graphical Report Prototype



63

63

Single Report Output Prototype

Member Order Detail

Order Information: 1

Order Number:	Promotion Number:	Member Number:
1929921	2	10001
Club Name:	Order Date:	Order Fill Date:
V	1/15/00	1/30/00
Order Status:	Sales Tax:	Pre-Payment:
	\$ 4.92	\$ 0.00
Sub-Total	Shipping Fee:	Amount Due:
\$ 98.58	\$ 3.50	\$ 107.00

Shipping Information: 1

Ship to Name:

Joe Smith

Shipping Address:

4589 Johnson Drive

City:

Lafayette

State:

IN

Zip Code:

47867

Shipping Instruction:

none

Navigation buttons: < << >> > Refresh Print Close

Flowchart: 1 (Order Information) -> 2 (Shipping Information) -> 3 (Navigation buttons)

64

64

Web Database Output

The screenshot displays the SoundStage Entertainment Club website. The left sidebar contains a 'Videos Search' section with a 'DTS Only' dropdown and a 'Browse by Category' dropdown set to 'Comedy'. The main content area features the product 'The Matrix 1999' with a 'Click box cover for preview' button. The product details include the SoundStage Price (\$18.99), Retail Price (\$24.98), and a savings of \$5.99. It lists the starring cast (Keanu Reeves, Laurence Fishburne, Carrie-Anne Moss, Hugo Weaving), the director (Andy Wachowski, Larry Wachowski), and features like 'Collector's Edition', 'Anamorphic Widescreen', '2.35:1 Aspect Ratio', 'Behind-The-Scenes', 'Commentary', 'Trailers', and 'Documentary'. The format is listed as 'DVD' with an MPAA Rating of 'R'. Below this, a table lists other available items:

Description	Format/Price	Purchase
Matrix, The (Widescreen)	DVD \$17.99	BUY
Matrix, The (Special Edition)	VHS \$14.49	BUY
Matrix, The (Special Edition)(Widescreen)	VHS \$14.99	BUY
Matrix, The (with CD Soundtrack, Poster, Lobby Cards) (Special Edition)	VHS \$67.99	BUY
Matrix, The: Musical Score From The Motion Picture, Music by Don Davis	CD Soundtrack \$11.99	BUY
Matrix, The: Music From The Motion Picture (EDITED)	CD Soundtrack \$12.99	BUY
Matrix, The - DVD and CD Soundtrack Set (Explicit Lyrics)	Other \$25.99	BUY

At the bottom of the page, there are links for 'Home', 'Privacy Statement', 'Store Locator', 'Help', 'Contact Us', 'Company Info', and 'Usb'. The page number '65' is visible in the bottom right corner.

65

Window Media Player Output

This screenshot shows the same SoundStage Entertainment Club website as the previous one, but with a Windows Media Player window overlaid. The media player window is titled 'SndStage.com - Preview Room - Microsoft Inte...' and 'The Matrix Preview'. It displays a video frame of Keanu Reeves talking on a mobile phone. Below the video frame is a standard media player control bar with buttons for play, pause, stop, previous, next, and volume. The website content is partially obscured by the media player window. The page number '66' is visible in the bottom right corner.

66

The Output Design Process

Step 1: Identify system outputs and review logical requirements

- Output requirements should have been defined during requirement analysis
- The data structure for a data flow specifies the attributes or fields to be included in the output.

67

67

The Output Design Process (cont'd)

Step 2: Specify physical output requirements

- Determine the best medium and format for the design and implementation based on
 - Type and purpose of the output
 - Operational, technical, and economic feasibility

68

68

The Output Design Process (cont'd)

- Is the output for internal or external use?
- If it's an internal output, is it a detailed, summary, or exception report?
- If it's an external report, is it a turnaround document?

69

69

The Output Design Process (cont'd)

Step 3: Design any preprinted forms

- What preprinted information must appear on the form? (contact information, headings, labels, and other common information to appear on all copied of form)
- Should the form be designed for mailing? If so address locations become important.

70

70

The Output Design Process (cont'd)

- How many forms will be required for printing each day? Week? Month? Year?
- What will be the form's size? Form size, along with volume can impact mailing cost.
- Will the form be perforated to serve as a turnaround document? If so, the location of the address become more critical.

71

71

The Output Design Process (cont'd)

- What legends, policies, and instructions need to be printed on the form (both front and back)?
- What colors will be used, and for which copies?

For external documents, carbon and chemical carbon are the most common duplicating techniques.

72

72

The Output Design Process (cont'd)

Step 4: Design, validate and test outputs

- Design the actual format of the report
- The best way to lay out the format is to sketch or, generate a sample of the report or document
- Show the sketch or prototype to the system user, get feedback, and modify the sample.

73

73

Design Issues

- Response time
- Help facilities
- Error handling
- Menu and command labeling
- Application accessibility
- Internationalization

74

74

Q&A

75

75